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# Mapping the Narrative Journey: A Structural Analysis of Podcast Conversations

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## Abstract

Podcast conversations unfold as dynamic sequences of interrelated ideas, yet most computational analyses treat them as static documents rather than evolving narratives. This project models the *narrative journey* of podcast episodes by analyzing how the narrative transitions over time. We develop an analytical pipeline combining state-of-the-art semantic segmentation with neural embedding, embedding-based topic labeling, and a set of statistical tools to capture the structural flow of long-form conversations. Using a small subset of the SPoRC (Spotify Podcast Corpus) [8] dataset, we examine genre-specific patterns in topic progression across health, education, and religion podcasts. Our work contributes a reproducible methodology and a set of visual tools for exploring the narrative journey of spoken dialogue.

## 1 Introduction

Long-form podcast conversations provide rich data for studying narrative evolution, yet traditional topic modeling often treats episodes as static text, overlooking their temporal dynamics. Understanding the structural flow of topics—such as the recursive or branching patterns found in different genres—is essential for characterizing communicative strategies [2]. However, despite the availability of large corpora, computational research examining these structural nuances remains limited.

In this project, we provide new insights into conversational structures, comparing journeys across different podcast genres (e.g., education, health and religion). The project delivers a methodology and a set of visualizations for studying the narrative architecture, semantic velocity, and temporal drift of long-form dialogue.

## 2 Related Work

Since their inception during the mid-2000s, podcasts have long been hailed as one of the most popular forms of media for consuming information across genres. Despite their popularity and reach, there has been surprisingly little large-scale research conducted until recently. From speech summarization [12], speech recognition [5], recommender systems [7] to identifying narrative content [1], podcasts were also analyzed for user engagement and popularity [10].

Our study, on the other hand, focuses on the structural analysis of the podcast conversation and how the narrative flows over time by leveraging state-of-the-art text segmentation techniques that improve upon the existing TextTiling [6] method, enabling us to quickly and efficiently analyze large amounts of text. This algorithm is the core of our pipeline. It leverages lexical frequency and data distribution to assess cohesion between different text segments, enabling it to identify sections likely to represent distinct topics based on these cohesion estimates as discussed in Unsupervised topic segmentation of meetings [13].

### 3 Dataset

We base our analysis on a subset (500 episodes) of SPoRC [8], which is a large-scale dataset of podcast transcripts spanning genres such as education, health, comedy, religion, news, sports and so on. SPoRC contains over a million episodes, each accompanied by metadata. For our purpose, we only needed these: 1. genre/category labels, 2. release date of the first episode, 3. full transcript text.

### 4 Methods

Our analytical pipeline consists of six components: (1) semantic segmentation, (2) embedding-based topic labeling, (3) topic-transition graph modeling, (4) visualization of the narrative flow, (5) semantic velocity or pacing, and (6) temporal drift, as explained below.

#### 4.1 Segmentation

The first stage of the pipeline divides each podcast transcript into coherent topical units. Segmentation quality affects downstream labeling and graph structure; thus we evaluated multiple algorithms before selecting a final method. We implemented and evaluated three segmentation families, each based on different assumptions about how topical coherence manifests in dialogue: **1. Fixed-Threshold Segmentation** **2. Adaptive Valley Detection** **3. Neural TextTiling**. Table 1 gives a brief comparison.

| Algorithm        | Core Idea                    | What it Measures                                              | How it Finds Breaks                                                               |
|------------------|------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------|
| segment_fixed    | <b>Grading on a Curve</b>    | Similarity between adjacent sentences ( $s_i$ vs. $s_{i+1}$ ) | Global threshold (e.g., 25th percentile). Breaks wherever similarity is below it. |
| segment_adaptive | <b>Finding Local Valleys</b> | Similarity between adjacent sentences ( $s_i$ vs. $s_{i+1}$ ) | Identifies local minima that are also lower than a scaled global mean.            |
| neuralTextTiling | <b>Comparing Landscapes</b>  | Similarity between blocks ( $k = 5$ before vs. $k = 5$ after) | Finds gaps where the preceding/following blocks differ most.                      |

Table 1: Comparison of Text Segmentation Algorithms

Since **neural TextTiling** provides the best result and scored high across different benchmarks according to the Unsupervised topic segmentation [13] paper, we select this as our primary segmentation algorithm. The implementation of our algorithm is based on the idea discussed in the paper. It produced the most consistent boundaries, was less sensitive to noisy or atypical sentences, and aligned qualitatively with human intuitions about topic changes in conversational speech.

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#### Algorithm 1: Neural TextTiling Segmentation (Simplified)

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Input: Sentences  $S_1 \dots S_n$ , embeddings  $E_1 \dots E_n$ , block size  $k$ 
    for  $i = k$  to  $n - k$  do
         $v \leftarrow$  mean embedding of  $E_{i-k+1} \dots E_i$ 
         $w \leftarrow$  mean embedding of  $E_{i+1} \dots E_{i+k}$ 
         $d_i \leftarrow 1 - \cos(v, w)$ 
    Smooth  $d$ ; identify depth valleys; insert boundaries at valley locations.
    return Segments

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##### 4.1.1 Sentence Embeddings

All segmentation methods rely on sentence-level semantic embeddings. Transcripts were first tokenized into sentences using the SaT model *sat-12l-sm* from **wtpsplit** [3] tokenizer. Each sentence

was then encoded using the *all-mpnet-base-v2* **SentenceTransformer** [11] model, which is optimized for semantic similarity tasks. Cosine similarity between adjacent embeddings served as the core coherence signal for local segmentation methods.

## 4.2 Topic Labeling

Once transcripts were divided into segments, we grouped each episode with respect to its genres. And since each segment required an interpretable topic label, for exploratory clustering and comparison, we adopted BERTopic [4] due to its high degree of flexibility when it comes to topic labeling. However, since segments vary in length and topical scope, and because our goal is to generate concise, human-readable labels for graph visualization, we implemented a custom tokenizer that filters n-grams based on part-of-speech patterns. This removes meaningless combinations and elevates the quality of the topic keywords. Finally, we integrated LLMs ( *gemma-3-4b-it-qat* [14]) running locally through an LM Studio API server with OpenAI’s client library to generate coherent titles for each topic.

## 4.3 Graph Modeling

To model the narrative flow, the sequence of topics was treated as a directed graph, where nodes represent distinct topics and edges represent chronological transitions. **Transition Weighting:** We calculated the frequency of every transition pair (i.e., how often *Topic A* leads directly to *Topic B*). These counts served as the edge weights, highlighting the "main paths" of conversation. **Noise Filtering:** We implemented a selective filter to remove "Outlier" topics (Topic -1). These outliers often act as artificial "hubs" in the graph, obscuring the true narrative structure by connecting unrelated themes.

## 4.4 Semantic Velocity and Pacing Analysis

While the narrative graph reveals the *structure* of a conversation, it does not capture its *density*. To quantify the "information pacing" of a podcast, we developed a metric termed **Semantic Velocity**.

**Vector Displacement:** We define Semantic Velocity ( $V$ ) as the Euclidean distance between consecutive segment vectors ( $S_t$  and  $S_{t+1}$ ) in the high-dimensional embedding space. A low velocity implies the speaker is dwelling on a concept (semantic closeness), while a high velocity indicates a rapid shift to a disparate topic.

**Archetype Classification:** By calculating the mean velocity ( $\mu_V$ ) and variance ( $\sigma_V^2$ ) for each episode, we classify podcasts into distinct pacing archetypes (e.g., "High-Density Scripted" vs. "Slow-Burn Conversational").

## 4.5 Longitudinal Analysis (Temporal Drift)

To analyze how the definition of a category evolves over years, we implemented a **Temporal Drift** analysis using the metadata timestamps associated with each episode.

**Centroid Tracking:** For each year  $Y$ , we compute the "Yearly Centroid" ( $C_Y$ ), defined as the mean vector of all segment embeddings belonging to that year. This represents the mathematical "center of gravity" for the genre at that specific time.

**Drift Quantification:** We measure the *Euclidean Shift Distance* between sequential years ( $\|C_Y - C_{Y-1}\|$ ) to quantify the magnitude of genre reinvention.

**Trajectory Mapping:** To visualize this evolution, we apply Principal Component Analysis (PCA) to project the high-dimensional centroids onto a 2D plane. This reveals the "Evolution Path" of the category, allowing us to detect cyclical trends, stabilizations, or sudden "pivot years" (e.g., the structural shift observed during the 2020 pandemic).

## 4.6 Evaluation

To determine the optimal segmentation for each podcast episode, we implemented a *Balanced Quality Score* metric. This evaluation framework guides an automated hyperparameter search over varying block sizes ( $k \in \{3, 5, 7, 9, 11, 15\}$ ), selecting the configuration that maximizes topical consistency while minimizing over-segmentation.

**Evaluation Metrics:** The quality of a segmentation is defined by the interaction of three primary components:

- **Intra-Segment Similarity ( $\uparrow$ ):** This measures the internal cohesion of a segment. We calculate the centroid of the segment’s sentence embeddings and measure the mean cosine similarity of all constituent sentences to that centroid.
- **Inter-Segment Similarity ( $\downarrow$ ):** This measures the distinctness between adjacent segments. A lower score indicates a more successful identification of a topical shift between  $Seg_i$  and  $Seg_{i+1}$ .
- **Fragmentation Penalty:** To prevent the trivial solution of single-sentence segments, we introduce a penalty based on the ratio of the number of segments to the total number of sentences.

**Scoring Function:** The final quality score  $Q$  is computed using the following objective function:

$$Q = (\mu_{intra} - \mu_{inter}) - (\lambda \cdot \frac{N_{segments}}{N_{sentences}}) \quad (1)$$

Where  $\mu_{intra}$  is the average intra-segment similarity,  $\mu_{inter}$  is the average similarity between adjacent segments, and  $\lambda$  represents the penalty weight (set to 0.2 in our implementation).

## 5 Result and Analysis

We divided the processed dataset of 500 episodes into three distinct genres—*Education*, *Religion*, and *Health*—to analyze their narrative structure, pacing dynamics, and longitudinal evolution.



Figure 1: Narrative flow of education, health, and religion categories with outliers from left to right respectively

### 5.1 Structural Topology

Initial graph modeling across all three genres revealed a consistent topological pattern: **Topic -1 (The Outlier)** acts as a universal "gravity well," (Figure 1) creating a dense "hairball" structure where distinct narratives collapse into a central hub of generic language. Filtering out this noise was essential to reveal the true underlying thematic architecture (Figure 2).

**Religion:** Removing outliers clarifies the structure into distinct theological concepts. For instance, a clear flow emerges from *Biblical Themes* into *Religious Concepts*, which remains structurally separate from the *Family Music Memories* cluster. This separation indicates that the category is not monolithic; rather, it segments clearly into "Scripture Study" episodes and "Personal Narrative" episodes.



Figure 2: Narrative flow of education, health, and religion with **no**-outliers from left to right respectively

**Education:** Similarly, the removal of the outlier hub fragments the graph into highly distinct thematic blocks. *Mental Health Concerns* emerges as a central node, while *Addiction Recovery* becomes an

isolated narrative arc. Interestingly, bridging topics appear, such as *Business People*, which connects disparate themes like *Air Force Career* and *Financial Rock Podcast*, highlighting the diverse, modular nature of educational content.

**Health:** In contrast to the other genres, the Health category displays a more fluid structure. Topics tend to flow rapidly within tight thematic loops. Even without outliers, the graph shows high interconnectivity, suggesting that health narratives shift topics more frequently and rapidly than education or religion.

## 5.2 Pacing Dynamics: Semantic Velocity Analysis

To quantify these structural observations, we applied Semantic Velocity analysis. This metric confirms that while all genres share some baseline characteristics, their pacing fingerprints are distinct (Figure 3).

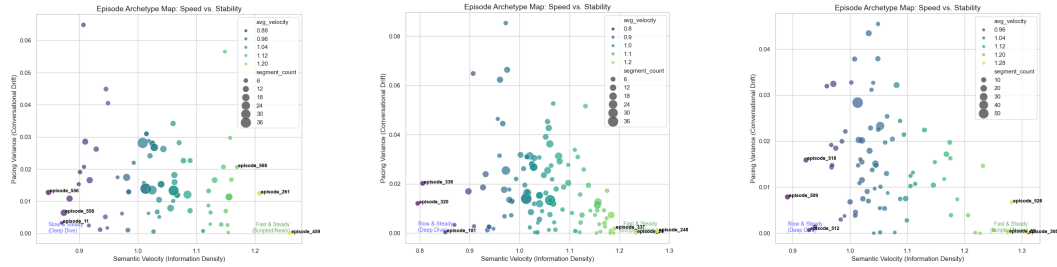


Figure 3: Episode archetype map of education, health, and religion categories (from left to right respectively) compares pacing vs stability across genre

**Education (Bimodal Structure):** This category exhibits the most diverse pacing. While the episodes are widely spread, there is a significant cluster of low-velocity episodes, indicating a funnel format where complex topics are elaborated upon slowly. This confirms that educational narratives are structured and deliberate, lacking the erratic conversational drift found in casual genres.

**Health (The "Goldilocks" Standard):** Unlike the fragmented Education category, the vast majority of Health podcasts follow a unimodal normal distribution. The pacing is "Goldilocks"—not as fast as a news briefing, but not as slow as a meditation. This is the mathematical signature of the Conversational Interview, balancing information density with listener engagement.

**Religion (Meditative Consistency):** The distribution for Religion is heavily skewed toward the lower velocity range. It exhibits slow, deliberate pacing with extremely low variance. Overall, Religion is the most structurally consistent genre analyzed, prioritizing stability and contemplation over rapid information transfer.

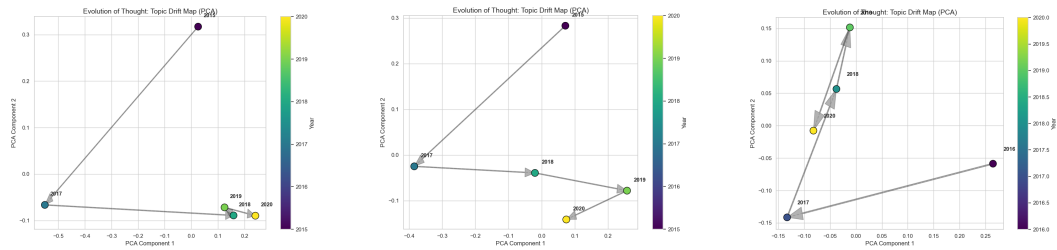


Figure 4: Temporal analysis of education, health, and religion categories (from left to right respectively) allows us to observe how the narrative evolved over time.

## 5.3 Longitudinal Evolution: Temporal Drift

Our Temporal Drift analysis (Figure 4) tracks how the semantic "center of gravity" for each genre shifted from 2015 to 2020. By correlating the topological movement in the PCA plots with the top

TF-IDF keywords extracted (Figure 5) for each year, we can explain the semantic drivers behind these shifts.

**Education (From Sociology to Self-Help):** This category underwent the most radical transformation. In 2015, the genre was defined by systemic social issues, dominated by keywords such as **"police," "violence," "Chicago,"** and **"officers,"** reflecting a focus on sociology and civic issues. The massive shift observed between 2017 and 2018 corresponds to a complete genre reformatting. The vocabulary pivots sharply from the systemic to the individual. By 2018, the lexicon is dominated by coaching and self-improvement terms: **"radical responsibility," "own choices," "coach,"** and **"boundaries."** This confirms that the "Education" category evolved from an academic/journalistic format into a personal development genre.

|   | 2015                 | 2017               | 2018                   | 2019         | 2020      |
|---|----------------------|--------------------|------------------------|--------------|-----------|
| 0 | police               | community colleges | own choices            | anxiety      | puffin    |
| 1 | poem                 | college futures    | gonna                  | rock         | puffin    |
| 2 | violence             | futures            | choices life           | gonna        | birds     |
| 3 | chicago              | higher education   | coach                  | confidence   | sea       |
| 4 | internationalization | colleges           | sharing                | assertive    | fun       |
| 5 | schools              | monica             | radical responsibility | air force    | stand     |
| 6 | affairs              | economic           | boundaries             | guitar piano | gonna     |
| 7 | freshhead            | futures foundation | our stories            | piano music  | addiction |
| 8 | erica                | workforce          | stories when           | career       | isle      |
| 9 | policing             | uc                 | marica                 | coffee       | isle may  |

|   | 2015            | 2017         | 2018                   | 2019                 | 2020         |
|---|-----------------|--------------|------------------------|----------------------|--------------|
| 0 | ted             | stellar      | sales                  | baby back            | south africa |
| 1 | ted head        | anchor       | choices                | re baby              | africa       |
| 2 | moreno          | podcasting   | own choices            | weight               | lockdown     |
| 3 | ted moreno      | rsi feed     | our stories            | calories             | virus        |
| 4 | head ted        | rsi          | linkedin               | diet                 | vegan        |
| 5 | hypnotherapy    | shares       | choices life           | fasting              | economy      |
| 6 | hypnotherapy    | wanna        | boundaries             | intermittent         | vaccine      |
| 7 | some truth      | upbeat music | teacher                | assertive            | discovery    |
| 8 | transformation  | re gonna     | another takes          | intermittent fasting | patients     |
| 9 | truth transform | hosting      | radical responsibility | um                   | inside covid |

|   | 2016          | 2017         | 2018      | 2019          | 2020           |
|---|---------------|--------------|-----------|---------------|----------------|
| 0 | yeshua        | il each      | police    | intelligence  | faith family   |
| 1 | ye            | other well   | uh        | ye            | holies         |
| 2 | wilson        | holy holy    | um        | cooper        | clerk          |
| 3 | thy           | open read    | jacobs    | organized     | uncommon       |
| 4 | poor needy    | read open    | braden    | sex           | stuck out      |
| 5 | bishop        | holy lord    | covenant  | intelligences | security       |
| 6 | never knew    | lord holy    | seattle   | blueberries   | uncommon sense |
| 7 | broken dreams | upbeat music | cops      | undisoptions  | bobby          |
| 8 | loves broken  | upbeat       | mormonism | healing       | wanna          |
| 9 | too busy      | music upbeat | fucking   | hall          | notice         |

Figure 5: Table contains the very exact topics (top 10) that evolved over time spanning five years across education, health, and religion categories (from left to right respectively)

**Religion (The Boomerang to Security):** The "Boomerang" trajectory observed in the PCA plot is explained by a return to fundamental themes during the crisis. Between 2018 and 2019, the genre drifted toward abstract and intellectual exploration, characterized by terms like **"intelligence," "organized,"** and **"healing"**. However, the sharp reversal in 2020 coincides with a vocabulary regression to safety and foundational dogma. The distinctive keywords for 2020 became **"faith family," "security,"** and **"notice."** This confirms that the pandemic forced religious content to abandon explorative theology in favor of providing immediate comfort and stability.

**Health (The Pandemic Pivot):** The hook-like trajectory in the Health category represents a forced thematic pivot within a stable structure. Up until 2019, the category followed a linear "Wellness March," defined by lifestyle optimization keywords such as **"intermittent fasting," "diet," "calories,"** and **"weight"**. The 2020 disruption did not change the format (interviews remained the norm), but completely overwrote the topic. The vocabulary of 2020 was instantly colonized by external events: **"lockdown," "virus," "vaccine," "economy,"** and **"South Africa."** This illustrates a reactive genre where the semantic content is dictated entirely by the global news cycle, unlike the evergreen nature of the pre-2020 wellness content.

## 5.4 Limitations

The paper has several limitations. For one, we only tested the pipeline on a fraction of the data (500 episodes of 1.1 million) so the performance will likely not generalize to the full dataset. Secondly, the categories in the dataset are not very accurate, making it very difficult to analyze them confidently. Similarly, as noted in the original paper of the dataset, Whisper [9] used for transcription is prone to hallucinations on non-majority accents, limiting the narrative diversity. Furthermore, our segmentation lacks optimal hyperparameter tuning due to computation cost.

## 6 Conclusion

This study successfully established a framework for analyzing the "narrative physics" of spoken audio, transcending traditional topic modeling by treating podcasts as dynamic, evolving structures. By integrating **Hybrid TextTiling Segmentation** with **Directed Graph Modeling**, we revealed that genres are defined not just by vocabulary, but by distinct structural architectures.

Longitudinally, our **Temporal Drift Analysis** uncovered a universal "genre reinvention" between 2017 and 2018 across all categories. Ultimately, this project demonstrates that structural embeddings can effectively "fingerprint" audio narratives, offering a novel, topology-based approach for content classification and recommendation systems.

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